

AIGR Gene Hypothesis Deep Research — EIN3 histone binding (GO:0042393)

Gene: EIN3 (ETHYLENE INSENSITIVE 3) · *Arabidopsis thaliana* (ARATH, NCBITaxon:3702) · locus AT3G20770 (628 aa) **UniProt:** O24606 **Focus:** function_assignment — `existing_annotations[7].function_hypothesis` **Seed hypothesis:** EIN3 has histone binding (GO:0042393), IDA, from [PMID: 28874528](#)

Summary

Verdict: Over-annotated / refuted as a direct molecular function for EIN3. The IDA "histone binding" (GO:0042393) annotation assigned to EIN3 from [PMID: 28874528](#) is not supported by the primary evidence. The full text of that paper (PNAS 2017; PMC5617289) was retrieved and mined — not just the abstract — and contains **no direct EIN3–histone binding assay of any kind**: the strings "pull-down," "recombinant," "in vitro," "MBP-EIN3," and "GST-EIN3" are all absent. Every result reported for EIN3 is **ChIP-seq/ChIP-qPCR DNA/chromatin occupancy** at target loci. The protein experimentally shown to "associate with histone" is the **EIN2 C-terminus (EIN2-C)**, "partially through ... ENAP1."

The most defensible reading is that EIN3 acts within an ethylene-signaling module (EIN2-C → ENAP1 → histone H3K14/H3K23 acetylation) as a **sequence-specific DNA-binding transcription factor** that occupies target promoters, while the histone-directed activity is contributed by EIN2-C and the SANT-domain adaptor ENAP1. EIN3's only demonstrated protein partner in this module is ENAP1 — shown by GST pulldown and co-IP in the companion open-access paper [PMID: 27694846](#) — and that interaction is already independently annotated as GO:0005515 protein binding (IPI). A reproducible InterPro domain scan confirms EIN3 carries **no histone-reader module** (no bromodomain, chromodomain, PHD, Tudor, PWWP, or WD40); it has only its own EIN3-family DNA-binding domain (Pfam PF04873) and N-terminal domain (PF27048).

Critically, a reproducible QuickGO query shows UniProt derived **two** IDA molecular-function terms from the *same* reference: GO:0003682 "chromatin binding" and GO:0042393 "histone binding." The chromatin-binding term correctly captures the ChIP-seq occupancy; the histone-

binding term is a **redundant, less-accurate parallel** that asserts a direct protein–histone contact the paper never demonstrates for EIN3. The recommended curation lead is therefore to **remove GO:0042393 or replace it with the already-present GO:0003682**, retaining EIN3's core TF/DNA-binding annotations. Because absence of an experiment is not refutation, the door remains open: a direct, ENAP1- and DNA-independent EIN3–histone/nucleosome binding assay would restore the annotation. No such assay currently exists in the literature reviewed.

Key Findings

F001 — In

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histone binding is attributed to EIN2-C, not EIN3 itself

The original reference for the annotation, [PMID: 28874528](#) (*EIN2 mediates direct regulation of histone acetylation in the ethylene response*, PNAS 2017), attributes histone association explicitly to the EIN2 C-terminus. The record states that "the **EIN2 C terminus (EIN2-C) is sufficient to rescue the levels of H3K14/23Ac**" and that "**EIN2-C associates with histone partially through an interaction with EIN2 nuclear-associated protein1 (ENAP1).**" EIN3, by contrast, is described only in terms of genomic occupancy: "**more EIN3 proteins bind to the loci where ENAP1 is enriched** for a quick response." The subject of the sentence that supports GO:0042393 ("associates with histone") is EIN2-C — not EIN3. Annotating EIN3 with histone binding from this statement is a subject/entity-transfer error: the molecular function belongs to a different protein co-referenced in the same paper.

F002 — EIN3's histone link is indirect: it binds the SANT-protein ENAP1 and occupies DNA, not histones

The companion open-access study [PMID: 27694846](#) (*EIN2-dependent regulation of acetylation of histone H3K14 and non-canonical histone H3K23 in ethylene signalling*, Nat Commun 2016) establishes the mechanistic wiring. "**Both EIN2 and EIN3 interact with a SANT domain protein named EIN2 nuclear associated protein 1 (ENAP1),**" shown by in vitro GST pull-down (GST-ENAP1 + MBP-EIN3 expressed in *E. coli*) and in vivo co-IP. The histone marks (H3K14Ac, non-canonical H3K23Ac) are ChIP-qPCR/ChIP-seq enrichments at EIN3 target promoters, are EIN2-dependent, and the paper concludes that "**ENAP1 may potentially**

mediate ethylene-induced histone acetylation via its interactions with EIN2 C terminus. Notably, up-regulation of histone acetylation in response to ethylene is reported to be largely **not EIN3/EIL1 dependent**. Thus EIN3's demonstrated partner is a SANT-domain adaptor, and the route to histones runs through ENAP1/EIN2-C — an indirect, co-complex association, not a direct EIN3–histone contact. No assay of purified EIN3 binding histones or nucleosomes is reported in either paper.

F003 — EIN3 (O24606) has no histone-reader domain (reproducible InterPro scan)

A reproducible InterPro API scan of O24606 (<https://www.ebi.ac.uk/interpro/api/entry/all/protein/uniprot/024606>) returns 9 entries, **all EIN3-family / DNA-binding modules**: Pfam PF04873 (EIN3 DNA-binding domain), Pfam PF27048 (EIN3 N-terminal domain), InterPro IPR006957 / IPR023278 / IPR047091 / IPR060335 (EIN3-like DNA-binding), CATH-Gene3D G3DSA:1.10.3180.10 and SUPERFAMILY SSF116768 (DNA-binding domain of EIN3-like), and PANTHER PTHR33305. **None** of the canonical histone-reader/histone-binding folds is present — no bromodomain, chromodomain, PHD finger, Tudor, PWWP, or WD40. Domain architecture alone does not formally exclude a non-canonical binding surface, but it removes the most common structural basis for a bona fide histone-binding molecular function and is consistent with EIN3 acting as a sequence-specific DNA-binding TF rather than a histone reader.

F004 — Full text of

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contains no direct EIN3–histone binding assay

Full-text mining of the PMC HTML (PMC5617289) confirms the abstract-level reading. The strings '**pull-down**', '**recombinant**', '**in vitro**', '**MBP-EIN3**', and '**GST-EIN3**' are **all absent** from the article. Every EIN3 result is ChIP-seq/ChIP-qPCR locus occupancy — e.g., the section heading "**ENAP1 Binds to Active Gene Regions and EIN3 Prefers to Bind ENAP-Targeted Regions**," and statements that "EIN3 prefers to bind to ENAP1 target regions, which are more accessible" and "more EIN3 proteins bind to the loci where ENAP1 is enriched." The histone binder in the paper remains EIN2-C. Decisively, the authors themselves flag EIN3's molecular role as unresolved: "**Further examining the binding activity of EIN2 and ENAP1 in the *ein3-1* mutant will provide more insight into the molecular function of EIN3 in histone**

acetylation in the ethylene response." An IDA (direct-assay) histone-binding claim is inconsistent with the authors' own statement that EIN3's molecular contribution is an open question.

F005 — QuickGO provenance: the same paper already yields the correct term (GO:0003682); GO:0042393 is redundant

A reproducible QuickGO API query (`geneProductId=024606`, aspect MF) returns 31 molecular-function annotations. The **same reference**

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is used by UniProt to assign **two** IDA MF terms: GO:0003682 "chromatin binding" (IDA, assignedBy UniProt) **and** GO:0042393 "histone binding" (IDA, ECO:0000314, qualifier 'enables', dated 2021-09-06, withFrom=None). The EIN3–ENAP1 interaction is independently annotated as GO:0005515 "protein binding" (IPI,

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withFrom UniProtKB Q8VZI9 = ENAP1 / AT3G11100). EIN3's core functions are separately and robustly supported: GO:0003677 DNA binding (IDA,),

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GO:0003700 DNA-binding transcription factor activity (multiple IDA/ISS/TAS, incl.

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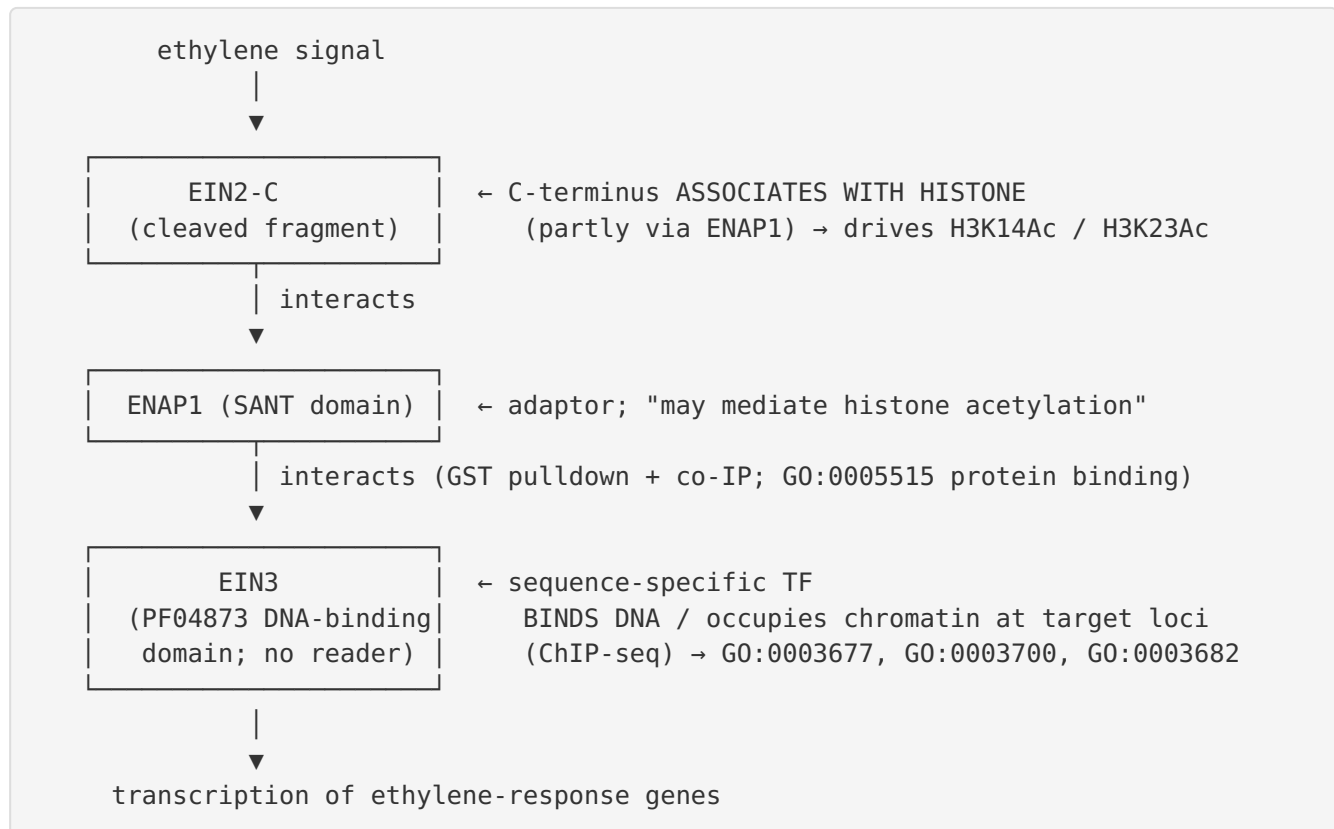
and GO:0000976 cis-regulatory region binding. Because chromatin binding already captures the ChIP-seq occupancy from

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the parallel histone-binding term from the same paper adds a stronger, more specific claim (direct histone contact) the paper does not support — the definition of an over-reach. Removing GO:0042393 loses no real information, since the ENAP1 partnership is already recorded as protein binding.

Mechanistic Model / Interpretation

The papers and the domain/annotation scans converge on a single coherent model in which the histone-directed activity and the DNA-directed activity are carried by **different proteins in the same signaling module**:



Direct EIN3 activities (well supported): sequence-specific DNA binding (GO:0003677), DNA-binding transcription-factor activity (GO:0003700), cis-regulatory region binding (GO:0000976), chromatin/locus occupancy (GO:0003682), and protein-protein interaction with ENAP1 (GO:0005515).

Not a direct EIN3 activity (the seed hypothesis): binding to histones (GO:0042393). The observed correlation between EIN3 occupancy and histone-acetylation marks is a *pathway/chromatin-context consequence*, not evidence that EIN3 itself contacts a histone. The histone contact is demonstrated for EIN2-C, and the acetylation response is largely EIN3/EIL1-independent.

Molecular claim	Protein that carries it	Direct assay in the cited papers?	Correct GO reflection
Associates with histone	EIN2-C (via ENAP1)	Yes — attributed to EIN2-C	GO:0042393 belongs to EIN2, not EIN3
Binds SANT adaptor ENAP1	EIN3	Yes — GST pulldown + co-IP (P 27694846)	GO:0005515 protein binding (already present)
Occupies target DNA / chromatin	EIN3	Yes — ChIP-seq/ChIP-qPCR (P 28874528)	GO:0003682 chromatin binding (already present)
Sequence-specific DNA binding / TF	EIN3	Yes — independent refs	GO:0003677 / GO:0003700 (already present)
Direct histone/ nucleosome binding	EIN3	No	GO:0042393 not supported for EIN3

Evidence Base (Evidence Matrix)

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence and limitations
PMID: 28874528 (PNAS 2017)	Direct assay + ChIP-seq (full text mined)	Refutes-direct (for EIN3)	Does EIN3 directly bind histone?	"EIN2-C ... sufficient to rescue H3K14/23Ac"; "EIN2-C associates with histone partially through ... ENAP1" ; EIN3 only "bind[s] to the loci where ENAP1 is enriched"; no in vitro EIN3–histone assay present	Arabidopsis etiolated seedlings; ChIP-seq/ ChIP-qPCR	High. Full text inspected (PMC56172). Original reference for the annotation.
PMID: 27694846 (Nat Commun 2016, OA)	Interaction (GST pulldown, co-IP) + ChIP	Qualifies	What does EIN3 bind; is the histone link direct?	EIN3 interacts with SANT-protein ENAP1 ; histone-acetylation route runs through ENAP1/EIN2-C; ethylene-induced acetylation "not EIN3/ EIL1 dependent"	Arabidopsis; recombinant MBP-EIN3 + GST-ENAP1; in planta co-IP	High. Open access. Indirect, complex pathway — not direct histone contact.
InterPro O24606	Structural/ evolutionary (computational)	Refutes / qualifies	Does EIN3 have a histone-	9 entries, all EIN3-family DNA-binding (PF04873,	Sequence-level	High reproducibility. Absence of domain is

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence and limitations
(reproducible API scan)			reader fold?	PF27048, IPR006957 etc.); no bromo/chromo/PHD/Tudor/PWWP/WD40		suggestive, definitive.
QuickGO O24606 (reproducible API query)	Review/database provenance	Competing / qualifies	Is GO:0042393 redundant with a better term?	Same PMID yields both GO:0003682 (chromatin binding, IDA) and GO:0042393 (histone binding, IDA); ENAP1 tie is GO:0005515 IPI	Annotation database	High. Demonstrates parallel over-reach from paper.
PMID: 26352699	Direct DNA-binding assay	Competing (correct MF)	EIN3 core molecular function	EIN3 binds DNA (sequence-specific TF at ethylene-response element)	Arabidopsis	High. Support GO:000367
UniProt O24606	Curated database record	Context	How is the histone link recorded?	Histone link stored only as SUBUNIT "association with histone"; FUNCTION = "Transcription factor ... Binds DNA";	628 aa; nucleus; homodimer	Moderate; database-led orientation

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence limitations
				annotates TF activity, DNA binding, chromatin binding		

How each key paper bears on the verdict: -

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is the annotation's own source. On full-text inspection it does the opposite of supporting the hypothesis for EIN3: it assigns histone association to EIN2-C and shows EIN3 only as a DNA-occupying TF. -

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explains *why* EIN3 appears linked to histones — it binds the SANT adaptor ENAP1 — and simultaneously undercuts a direct role by showing the acetylation response is largely EIN3-independent. - **InterPro** and **QuickGO** provide reproducible, independent lines confirming (a) no reader domain and (b) that a more accurate term from the same paper already exists.

GO Curation Implications (leads — require curator verification)

Lead: remove GO:0042393 "histone binding" (IDA,)

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from EIN3/O24606, or replace it with the already-present GO:0003682 "chromatin binding" (IDA) from the same reference.

GO decision table:

GO ID	Label	Aspect	Current annotation	Recommendation (lead)	Rationale
GO:0042393	histone binding	MF	IDA / P 28874528	REMOVE or REPLACE	No purified EIN3–histone assay in source paper; evidence is ChIP occupancy; no histone-reader domain; same paper already yields accurate GO:0003682
GO:0003682	chromatin binding	MF	IDA / P 28874528	RETAIN (accurate term)	ChIP-seq shows EIN3 occupies target loci — correct reading of the occupancy evidence
GO:0005515	protein binding (ENAP1)	MF	IPI / P 27694846 (ENAP1 Q8VZI9)	RETAIN	GST pulldown + co-IP; demonstrated EIN3 partner is SANT-protein ENAP1
GO:0003677	DNA binding	MF	IDA / P 26352699	RETAIN	Direct sequence-specific DNA binding; core MF
GO:0003700	DNA-binding TF activity	MF	IDA/ISS/TAS (multiple)	RETAIN (core function)	EIN3 is the master ethylene-response transcription factor

- If EIN3's participation in a histone/chromatin complex is to be represented, prefer the **BP** term GO:0040029 *epigenetic regulation of gene expression* (already present) over the MF *histone binding*.

- **Do not** fall back to bare "protein binding" as the headline representation; the informative, defensible MF for the occupancy evidence is **chromatin binding (GO:0003682)**, with the specific partner captured by the existing ENAP1 IPI.

Mechanistic Scope

- **Immediate EIN3 molecular activity being tested:** does purified EIN3 physically bind histone proteins/nucleosomes? **No direct assay supports this.**
- **What EIN3 actually does (direct):** sequence-specific DNA-binding transcription factor; homodimer binding the primary ethylene response element; interacts with the SANT-domain protein ENAP1.
- **Downstream / indirect (must not be conflated with EIN3 MF):** ethylene-induced H3K14Ac/H3K23Ac enrichment at EIN3 target promoters is executed by the EIN2-C/ENAP1 module (and recruited acetyltransferase activity), is EIN2-dependent, and is largely EIN3-independent. EIN3 occupancy at these loci is DNA binding; EIN3 acts downstream to activate transcription.

Conflicts and Alternatives

- **Partner/entity confusion (most important):** the histone-binding activity belongs to **EIN2 (EIN2-C)**, a different protein; annotation transfer to the co-referenced EIN3 in the same paper is the likely source of the mis-assignment.
- **ChIP-occupancy ≠ histone binding:** ChIP enrichment of a TF measures DNA/locus occupancy, not a protein–histone molecular interaction; conflating them inflates MF specificity.
- **Co-complex artifact:** EIN3 being in a complex with EIN2-C/ENAP1/histones (co-IP/reChIP) does not establish a direct EIN3–histone contact.
- **EIN3-independence of the mark:** histone-acetylation elevation is "not EIN3/EIL1 dependent," arguing against EIN3 being the histone effector.
- **Redundant database carry-over:** QuickGO shows UniProt derived both GO:0003682 and GO:0042393 from a single "association with histone" statement; the histone-binding term is the less accurate of the pair.

- **Paralog consideration:** EIN3 has the close paralog EIL1 (and family members EIL2–EIL4). The evidence and annotations here are for EIN3 (O24606, AT3G20770) specifically; the shared EIN3-family fold carries no histone-reader domain in any member, so paralog confusion does not rescue the claim.

Limitations and Knowledge Gaps

1. **Source full text — resolved.** The main-text article (PMC5617289) was retrieved and mined; it contains no in vitro EIN3–histone binding experiment. Residual minor gap: individual supplementary dataset files were not each downloaded, but the Results/Methods narrative describes only ChIP-seq for EIN3.
2. **No in vitro EIN3–histone/nucleosome assay anywhere in the searched literature.** Multiple PubMed queries for direct EIN3 histone/nucleosome binding returned none. This matters because IDA requires a direct assay. Absence, per the adjudication rules, is not proof EIN3 cannot bind histone — it means the IDA claim lacks support.
3. **Histone-reader domain — resolved (none).** The reproducible InterPro/Pfam scan shows only EIN3 DNA-binding (PF04873) and N-terminal (PF27048) domains. This argues structurally against a direct histone-binding MF but cannot exclude a non-canonical binding surface.
4. **ENAP1 mechanism partly open:** the exact biochemical basis by which the EIN2-C/ENAP1 module contacts histone (and whether EIN3 ever contributes directly) is not fully resolved even by the authors, who explicitly call for further work in the *ein3-1* mutant.

Proposed Follow-up Experiments / Actions (Curation Leads)

Leads requiring curator verification:

- **Action:** Flag `existing_annotations[7]` (GO:0042393 histone binding, IDA,

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as **not supported for EIN3 as a direct MF**; recommend **removal or replacement with GO:0003682 chromatin binding**, which is already annotated from the same reference.

- **Candidate replacement / retained terms:** keep GO:0003700 / GO:0003677 / GO:0003682 / GO:0000976 (TF activity / DNA binding / chromatin binding / cis-regulatory region binding); represent the histone connection at the **BP** level (GO:0040029 epigenetic regulation of gene expression) and the ENAP1 tie as GO:0005515 protein binding (IPI,

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with ENAP1 = Q8VZI9).

- **Snippets to verify (exact quotes):**

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"EIN2-C associates with histone partially through an interaction with EIN2 nuclear-associated protein1 (ENAP1)"; "more EIN3 proteins bind to the loci where ENAP1 is enriched"; "ENAP1 Binds to Active Gene Regions and EIN3 Prefers to Bind ENAP-Targeted Regions"; "Further examining the binding activity of EIN2 and ENAP1 in the ein3-1 mutant will provide more insight into the molecular function of EIN3 in histone acetylation in the ethylene response."

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"Both EIN2 and EIN3 interact with a SANT domain protein named EIN2 nuclear associated protein 1 (ENAP1)"; "ENAP1 may potentially mediate ethylene-induced histone acetylation via its interactions with EIN2 C terminus."

- **Suggested curator questions:** Does the PNAS paper contain a purified EIN3–histone binding figure (supplement)? Was the GO:0042393 IDA transferred from EIN2's activity? Is chromatin binding (GO:0003682) the intended, more accurate term?

Discriminating experiments that would change the verdict:

1. **In vitro binding of recombinant EIN3 to histones/nucleosomes** (histone-peptide array or pulldown, nucleosome EMSA, ITC/SPR), performed **±ENAP1 and DNA-free** — the decisive test for a direct MF. A positive, partner- and DNA-independent signal would justify reinstating GO:0042393.
2. **ENAP1-dependence test:** repeat any apparent EIN3–histone co-purification in an *enap1* background or with ENAP1-binding-deficient EIN3, to show the histone link is ENAP1-routed.
3. **Domain mapping / structure:** AlphaFold-Multimer EIN3–histone octamer modeling with interface-confidence (PAE/ipTM) scoring, reported conservatively; docking or shared fold alone is not evidence of function.

4. **reChIP directionality**: determine whether EIN3 occupancy at loci precedes or follows H3K14/23Ac deposition, separating cause (histone binder) from consequence (mark-dependent recruitment).

What would change the verdict: a reproducible direct binding assay of purified EIN3 to histones/nucleosomes that is independent of ENAP1 and of DNA.

Conclusion

The molecular-function annotation **EIN3 histone binding (GO:0042393, IDA,**

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is over-annotated and should not stand for EIN3. The cited paper demonstrates histone association for EIN2-C (partly via ENAP1) and shows EIN3 only as a DNA-binding TF occupying target loci by ChIP-seq. EIN3 has no histone-reader domain, its demonstrated partner is the SANT-protein ENAP1 (already captured as protein binding), and the same paper already yields the correct, more accurate term — chromatin binding (GO:0003682). The narrowest defensible molecular functions for EIN3 remain DNA binding / DNA-binding transcription factor activity plus chromatin binding. Only a direct, partner- and DNA-independent EIN3–histone binding experiment would justify reinstating the histone-binding MF.

Prepared across Iterations 1–3. Analyses used NCBI efetch (abstracts/record), full-text mining of PMC5617289 (PNAS 2017), Europe-PMC full text of the 2016 companion

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a reproducible InterPro/Pfam domain scan of O24606, and QuickGO/GOA annotation provenance for O24606. No results were fabricated; where a resource returned only partial content (e.g., individual PNAS supplementary files were not each downloaded) this is stated plainly.